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Research paper

A cuckoo load scheduling optimization approach for smart energy management

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ABSTRACT

Several load scheduling optimization models have been proposed for smart grid systems. Nevertheless, most of the existing models assume the presence of a pricing mechanism such as time of use (TOU) and consider single objective functions that are suitable for application under specific conditions. This paper proposes a generic load scheduling optimization model that can be employed under different operating conditions and can handle different scheduling criteria. In particular, the model considers TOU and allows the integration of distributed renewable energy systems (DRES). In addition, the load scheduling model is solved using the Cuckoo optimization algorithm. Performance of the Cuckoo algorithm is validated by formulating and solving an equivalent MILP model to the load scheduling problem. A set of experiments is designed to compare optimality and time performance of the Cuckoo and its equivalent MILP model. The Cuckoo results has also shown superior or at least comparable results to the published results in the literature. A case study has been performed using real data taken from an academic building in Egypt to demonstrate the model applicability under different conditions. The results show that the building under specific conditions can achieve energy cost savings that range from 57% to 80%. The results have also provided useful managerial implications.

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1. Introduction

Most of the existing power grids are inefficient and outdated and need to be upgraded into smart grids. A smart grid is an integration of electric power grid with bidirectional communication network system. It integrates three major subsystems: smart infrastructure subsystem, smart energy management subsystem, and smart protection subsystem (Latifi et al., 2017; Abdelhady et al., 2020). Demand Side Management (DSM) is an essential function in smart grids for enabling the smart energy management. Usually, the main goal of DSM is to encourage consumers to consume less power during peak times or to shift their energy use to off-peak times in order to flatten the load profile. It can be used to shift loads over time in order to better match the demand and supply of electric energy, instead of enlarging the generation capacity. In addition, DSM allows for better integration of renewable energy systems that are characterized by its intermittent nature. It can also be integrated with energy saving measures to achieve the energy and economic goals (Tang et al., 2020; Calise et al., 2021).

DSM has received a significant attention in smart grid research (Latifi et al., 2017; Esther and Kumar, 2016; Monfared et al., 2019). There are various DSM approaches such as load shifting, peak clipping, conservation, load building, valley filling and flexible load effect (Esther and Kumar, 2016; Gelazanskas and Gamage, 2014). In particular, several studies have been devoted to load scheduling which can be applied through load shifting to alter load profile at consumer premises (Lezama et al., 2020a). The main incentive which is always assumed to engage the consumers in DSM and allow load shifting is the time of use pricing mechanism (TOU) which is commonly adopted in developed countries. Accordingly, most of the existing load scheduling models are mathematically formulated with assuming the presence of an incentive mechanism such as TOU. However, several countries (especially developing countries) do not rely upon TOU for DSM, therefore, the existing load scheduling models are not suited for them. As such, engaging the consumers of those countries in such DSM programs is not feasible unless other benefits can be realized. A possible incentive is to allow the consumers that have installed distributed renewable energy systems (DRES) such as PV systems to effectively utilize the output of their DRES such that they can schedule their loads in the manner that utilizes the power output and thus minimizing the reliance on the grid (Lezama et al., 2020b). As such, the cost paid by the consumers to the grid can be minimized through load scheduling to better match the output of installed DRES. Also, regardless of the incentive mechanism, most of the existing load

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scheduling models have been considering single objective functions (scheduling criteria) such as cost minimization or peak load minimization. Accordingly, load scheduling models that consider different single/multiple scheduling criteria have not received a significant attention (Foroozandeh et al., 2021; Deng et al., 2015; Chavali et al., 2014). As such, the existing load scheduling models have a limited applicability and thus it should be adapted to allow the effective utilization of DRES, and the possibility to address the load scheduling problem based on different scheduling criteria, while considering the system constraints.

This paper proposes a generic load scheduling optimization model that can be applied under different operating conditions and can be applied to optimize load schedules based on different scheduling criteria. The model takes into account the existence of TOU pricing mechanisms, and allows the integration of DRES. The load scheduling model is mathematically formulated such that it can be adapted to handle different preferences from both consumer and grid sides, while taking into account the existence of TOU, DRES, or both. In addition, combinations of the scheduling criteria (cost, peak load, user inconvenience) can be considered to address the load scheduling from different perspectives. As such, the load scheduling model can be applied for energy management in smart grid systems of different countries or communities. In particular, it can be applied for energy management of the households, residential and commercial buildings. The cuckoo optimization algorithm is used to solve the load-scheduling problem as it has shown a promising performance in previous related research (Laha and Gupta, 2018). The integration of the proposed load scheduling model with the cuckoo optimization algorithm is called in this paper as the cuckoo load scheduling approach.

Several experiments have been performed to validate the performance of the cuckoo load scheduling approach, and to demonstrate its usefulness for real-life applications under different operating conditions. In particular, a mixed integer linear programming (MILP) model is formulated for the same load scheduling problem for validating the optimality and time performance of the cuckoo load scheduling approach. The MILP model is only used for validation of the Cuckoo solution optimality. It is not intended as a solution methodology due to the exploding problem sizes and computational time for realistic size problems. The validation results indicate that the cuckoo provides a promising performance compared to the MILP-based load scheduling approach. Moreover, the performance of the cuckoo approach has outperformed the published results in literature (AboGaleela et al., 2012; Vidal et al., 2014). Finally, a case study is undertaken using real data taken from an academic building in Egypt. The results show that the building, under specific conditions, can achieve energy cost savings that range from 57% to 80% through applying the proposed load-scheduling model. The results confirm the applicability of the proposed load scheduling model in the situations where TOU is not applicable as in the residential sector of Egypt. This proves the wide applicability of the proposed load scheduling model compared to the existing models in the literature.

The main contribution of this paper spans the proposed load scheduling model that can be applied for energy management in modern and emerging smart grids. The model is adapted from the existing models in the literature where some modifications have been made to allow the model to be widely applicable. The proposed mathematical model can be adapted in future research to consider other aspects of the load scheduling problem in smart grids. Moreover, the integration of the cuckoo optimization algorithm with the proposed load scheduling model has provided a promising performance compared to both the MILP-based load scheduling approach and the relevant results in the literature. The reported results can be used for benchmarking other future formulation approaches and solution algorithms. The results analysis has also provided useful managerial implications.

The paper is organized as follows. The literature review is discussed in Section 2. The system description is presented in Section 3. The optimization approach is presented in Section 4. The validation results are introduced in Section 5, and the case study is presented in Section 6. The managerial implications are discussed in Section 7. The conclusions and future research direction are presented in Section 8.

2. Literature review

Extensive research has been directed to the analysis and evaluation of different DSM approaches within the context of smart grid systems (Tang et al., 2020; Calise et al., 2021; Esther and Kumar, 2016; Monfared et al., 2019; Gelazanskas and Gamage, 2014; Lezama et al., 2020a,b; Chakir et al., 2020). This literature review focuses on the previous DSM studies that are related to the integration of renewables in smart grid through DSM, load scheduling problem, and optimization approaches for the load-scheduling problem.

DSM is a smart grid function that aims at matching energy demand with available supply through altering load profile at consumer premises. Moreover, the widespread of DRES necessitates the presence of intelligent energy management systems to manage load demand considering the intermittent nature of renewables (Chakir et al., 2020). Therefore, several studies have focused on the integration of renewables and storage systems in smart grids through DSM (Wang et al., 2015). Adika and Wang (2014) introduces an intelligent energy management framework that can consider both energy storage and appliance scheduling schemes to flattening the load profile. Power storage allows consumers to purchase power during off-peak hours and satisfy their demands at peak by discharging the batteries. Melhem et al. (2017) proposed a load scheduling model to minimize the electricity bill of the residential consumers. The model considers the integration of renewable energy production, and battery storage system. Latifi et al. (2017) proposed a demand side management model to minimize consumer utility costs and maximizes comfort levels in a fully distributed manner. The model allows consumers to sell excess electricity (e.g., from renewable resources PV or wind turbine) back to the grid in order to reduce their electric utility bill. Liu (2016) investigated the problem of minimizing electricity cost and the peak load in smart grids with distributed renewable energy resources and assumed that all consumers are equipped with a solar-plus-battery system. Han et al. (2018) investigated a fast randomized first-order optimization method to explore the solution of dynamic energy management for the smart grid integrated large-scale distributed energy resources. Their proposed optimization method considers different demand response programs and schedules both elastic load and inelastic load, as well as the operations of distributed renewable energy resources and energy storage devices. Most of the previous studies have been relying on load scheduling to realize the efficient integration of DRES.

Load scheduling is the most promising DSM approach in which the consumer shifts loads from peak hours to off peak hours to minimize electricity bill cost and grid power peaks. In this regard, extensive research has been devoted to the application of load scheduling as a DSM approach (Zhang et al., 2016; Nojavan et al., 2018; Haider et al., 2016). Bradac et al. (2015) proposed a scheduling model to minimize the total energy cost paid by the consumer and to reduce power peaks in order to achieve a balanced daily load schedule. An exact and computationally mixed-integer linear programming (MILP) formulation is used to

determine the optimal load schedule. Hassan et al. (2013) investigated the impact of load shifting flexibility on both demand profile flatness and consumer inconvenience in residential smart grid systems. Gottwalt et al. (2011) developed a DSM model using load shifting that generates realistic electricity load profiles that are close to empirically measured profiles.

Different solution approaches are proposed to tackle the DSM load scheduling problem. Some researchers used linear programming and mixed integer linear programming in order to minimize bill cost or peak load. Manzoor et al. (2017) proposed a hybrid teacher learning genetic optimization technique (TLGO) by combining genetic algorithm (GA) with teacher learning based optimization (TLBO) algorithm for residential load scheduling. The major focus of their work is to minimize consumer electricity bill at minimum user discomfort. The performance of heuristic techniques is compared with LP in terms of cost, peak power consumption, peak to average ratio (PAR) and user discomfort. Shakouri and Kazemi (2017) proposed a multi-objective mixed integer linear programming model that can be used to minimize both electrical peak load and electricity cost for a residential area with multiple households. Chai et al. (2015) proposed a hybrid mixed-integer quadratic optimization model to optimize three indices: the power consumption expense, the robustness of schedule subject to uncertain electricity price, and the satisfaction of customer which results in a hybrid mixed-integer quadratic optimization problem. Foroozandeh et al. (2021) formulated a multi-objective mixed-binary linear programming that minimizes the total energy consumption cost and peak load, considering residential buildings, electric vehicles and battery energy storage system. Several other studies resorted to the MILP formulation of the problem (Melhem et al., 2017; Shakouri and Kazemi, 2017).

With various operating constraints, load scheduling is a complex optimization problem and the current classic load scheduling techniques do not lead to optimal solution in complex cases. Therefore, metaheuristics have been investigated widely with the load scheduling problem (Manzoor et al., 2017; Remani et al., 2015). Many metaheuristics-based scheduling algorithms are proposed in the literature for DSM load shifting problem considering different objective function and constraints. Recently, several studies have proposed and evaluated several techniques such as Particle Swarm Optimization (PSO) algorithm (Lezama et al., 2020a; Access et al., 2015; Gupta et al., 2016), Evolutionary and Genetics Algorithms (Logenthiran et al., 2012; Bharathi et al., 2017), and MIP-and-refine metaheuristics (Fischetti et al., 2015). Vidal et al. (2014) proposed an evolutionary algorithm to solve the load scheduling problem from two perspectives: cost minimization and peak load minimization. They have tested the effectiveness of their proposed approach through simulating a smart grid that consists of three areas: residential, commercial and industrial. Javaid et al. (2017a) developed a hybrid genetic wind-driven (GWD) algorithm for a residential area to maximize consumer comfort and minimizing both electricity cost and the peak to average ratio. Della Croce et al. (2017) discussed the problem of scheduling home appliances within a given time horizon respecting the threshold on the energy peak consumption with the objective of minimizing the overall electricity cost related to the average energy consumption of each appliance at each time. They proposed a hybrid heuristic based on a mixed integer programming formulation of the problem and compared it with a particle swarm optimization approach. Lezama et al. (2020b) compared different evolutionary algorithms (vortex search, differential evolution and particle swarm optimization) in solving the load scheduling problem, considering two different optimization approaches: parallel approach and non-parallel approach. Their results indicate that the parallel approach improves the performance of the adopted algorithms for large problem sizes.

Most recently, a few studies have applied cuckoo search algorithm for load scheduling problems in different applications, including the DSM application. They have shown that cuckoo search can realize a comparable performance to other known metaheuristics (Lakshminarayanan and Kaur, 2018; Laha and Gupta, 2018). Lakshminarayanan and Kaur (2018) have investigated the generator maintenance scheduling problem and found that the discrete cuckoo algorithm outperforms the genetic algorithm, discrete particle swarm optimization, and their modified versions. Cakmak and Altas (2016) proposed an approach using Cuckoo search algorithm that can be employed to increase the efficiency and lifetime of utility network by scheduling shiftable domestic loads to ensure balanced load curve as much as possible. Khalid et al. (2018) schedules home appliances using cuckoo search considering the objective of reducing the bill cost and peak load. Javaid et al. (2017b) compared cuckoo search with GA and BPSO to generating optimum schedules. Their model aims at optimizing the operation time of electrical appliances taking into account the amount of renewable energy generated and cuckoo search provide better results in comparison to GA and BPSO.

3. System description and formulation

3.1. System overview

We consider a smart grid system that contains K consumers (residential units, commercial units or both). Each consumer unit k ($k = 1, \ldots, K$) has a set of loads/appliances, and is supplied with energy from the power grid and a distributed renewable energy system (see Fig. 1). Each household has a smart meter which can also be connected to a Centralized Scheduler (CS). The smart meter gathers all the required information about consumer loads and communicates it to CS. The scheduling algorithm is executed by CS and the resulting load shifting values, consumer bill cost, peak load and user inconvenience levels are also measured by CS and are communicated to the power grid operator. Moreover, the smart meter and local controllers can be programmed to optimize load schedules of each consumer in a decentralized manner in order to address the scalability problem (Korkas et al., 2016). Furthermore, the smart meter can communicate the loads data to CS which in turn optimizes the load schedules for each user individually, considering one or more scheduling criteria. Then, the CS communicates the load schedules to the smart meters of the users to implement it. An illustrative diagram of the energy management system is provided in Fig. 1.

In particular, the problem can be described as follows. A set of K consumers that each consumer has a set of flexible loads/appliances M_k that must be scheduled within a defined time horizon T, i.e., determining the starting times of loads/appliances based on some scheduling criteria. The scheduling time horizon T is partitioned into time slots of fixed duration τ (time unit), where $t = 1, \ldots, T$. Each consumer's flexible appliance i, where $i = 1, \ldots, M_k$, has a duration (service time) $d_{i,k}$, a preferred time $r_{i,k}$ and once switched on, must work without interruption consuming amount of power, $p_{i,k}$.

In this system, it is assumed that each consumer has a PV system that its power output in each time slot t can be predicted a day ahead. The aggregate amount of the generated power represents the target load profile that the consumers' consumption should match to minimize the requested power from the grid. Then-after, the cost savings can be shared among the consumers. The PV output prediction is an important input for the efficient management of energy supply and demand in smart grids (Marzband et al., 2018). In particular, the prediction of PV output is essential to develop load schedules that effectively utilize the availably energy supply from such renewable systems

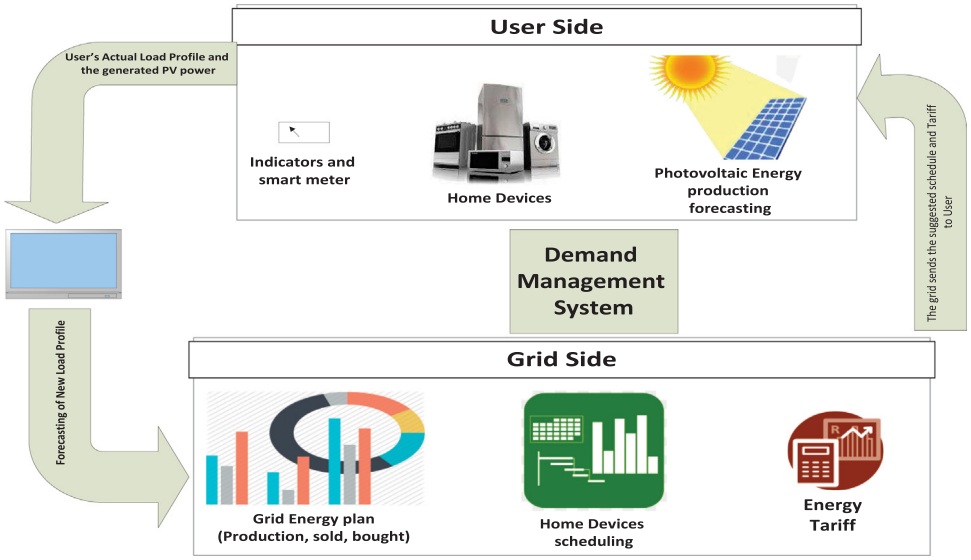


Fig. 1. A schematic diagram of the energy management system.

that are characterized by its variable power output (Haider et al., 2016). In this regard, extensive research has proposed different forecasting models, such as time series models, causal models and artificial intelligence models, that can be employed to accurately predict PV output (Raza et al., 2016; Zhou et al., 2019). Most importantly, there are several forecasting methods that can provide accurate predictions of the PV generation for one day ahead (Nespoli et al., 2019; Yang et al., 2014). As such, we assume that the grid/consumers have advanced forecasting systems that can predict the power output in each time slot for a day ahead. The uncertainty of forecasted PV output can be absorbed by energy storage systems or connecting with the power grid. In particular, we assume that the consumers are supplied with both the grid and PV systems such that the time periods with insufficient PV output to satisfy the load will be substituted from the grid. The PV output that exceeds the load demand will be injected into the grid. Furthermore, we assume that the consumers do not have storage systems to store the excess energy production of the PV systems. In particular, this assumption is justified when the installed PV is directly connected to the power grid (on-grid system) such that the excess PV output is directly injected into the power grid and be imported when it is needed. This assumption is also justified by the high cost of storage systems and that low-income communities will not be able to afford such systems. Some studies have considered this assumption while developing their load scheduling models (Della Croce et al., 2017).

In this case, the problem that faces the consumers is how to schedule the starting times of their loads in a way that maximizes the utilization of aggregate power output of the PV systems such that the total bill cost of the consumers is minimized. This requires the determiniation of the shifting values for consumers flexible load ($u_{i,k}$) that minimizes the consumer's bill. The scheduling can also be developed based on other criteria either single or multiple scheduling criteria.

3.2. Load types and assumptions

According to consumer usage and load function, loads can be categorized mainly into two categories: non-shiftable and shiftable loads. The shiftable loads can be scheduled to constitute balanced load curve as much as possible, considering the preferences of the consumers. Non-shiftable appliances/loads (or Essential loads) have a fixed scheduling over the time horizon and cannot be re-scheduled or shifted. For instance, refrigeration, watching television (TV), essential lighting are a few examples of such essential loads. Time-shiftable appliances/loads (or flexible loads) have preferred scheduling times defined by customers, and these loads can be re-scheduled to other time slots over the time horizon.

The DSM is carried out at the beginning of a predefined planning period which is typically a day. The time during a day is partitioned into T time slots, indexed by t ($t = 1, \ldots, T$). The duration of each time slot, denoted by τ, is fixed where it can be an hour, half hour, etc. Each consumer k has a set of non-shiftable appliances, where $N_k(t)$ represents the power consumption of non-shiftable loads of consumer k in time slot t. The total power consumption of non-shiftable loads in time slot t, $N(t)$ is obtained from the summation of non-shiftable loads demand/consumption in period t as depicted in Eq. (1) (Hassan et al., 2013).

$$N(t) = \sum_{k=1}^{K} N_k(t), \forall t \tag{1}$$

In addition, each consumer k has M_k flexible loads or appliances, $d_{i,k}$ is the duration (service time) of flexible load i of consumer k. We assume continuous operation of each flexible load until its completion. It means that if the flexible load is scheduled at time slot $t = \tau$ then it cannot be stopped and it will operate in consecutive time slots $\tau, \tau + 1, \ldots, \tau + d_{i,k} - 1$.

Total energy consumption at time slot t is obtained from the summation of non-shiftable and flexible loads scheduled at period t as shown below (Hassan et al., 2013):

$$SG(t) = N(t) + \sum_{k=1}^{K} \sum_{i=1}^{M_k} e_{i,k}(t), \forall t \tag{2}$$

where $SG(t)$ expresses the total consumption at time slot t, and $e_{i,k}(t)$ is the electricity consumed by a flexible load i of consumer k at time period t. The power consumption $e_{i,k}(t)$ is assumed to be fixed and equal to $p_{i,k}$ over the service time as expressed in Eq. (3) (Hassan et al., 2013).

$$e_{i,k}(t) = \begin{cases} p_{i,k}, & \forall t = \tau, \tau + 1, \ldots, \tau + d_{i,k} - 1 \\ 0, & otherwise \end{cases} \tag{3}$$

Each flexible load is assumed to have a preferred scheduling time (start time), denoted by $r_{i,k}$, that is defined by the consumer. The load i of consumer k can be characterized by four parameters (duration, power consumption, preferred scheduling time and shifting slots allowed). Thus, we can refer to the load data by the following notation, $F_{i,k} = [d_{i,k}, p_{i,k}, r_{i,k}, X_{i,k}]$.

3.3. Load shifting

The flexible/shiftable loads can be scheduled with delay or advancement depending on the requirements of the grid and the permissible consumer inconvenience level. The consumer provides the grid with the preferred time for running each flexible load and the permissible period that can be shifted earlier than or beyond the preferred time. Scheduling of a flexible load can be done on either side of its preferred time slot, e.g., clothes drying can be scheduled earlier than consumer's preferred time or can be delayed. In general, each flexible load $F_{i,k}$ can be shifted by $X_{i,k}$ time slots. Thus, each flexible load $F_{i,k}$ can be started within the time interval, $[r_{i,k} - X_{i,k}, r_{i,k}+X_{i,k}]$. Since all loads have to be scheduled within the time interval $[1, T]$, the starting time of each $F_{i,k}$ has also to be bounded by the time interval $\left[\tilde{L}_{i,k}, \tilde{U}_{i,k}\right]$ that can be determined as in Eqs. (4)–(5).

$$\tilde{L}_{i,k} = \max\left(1, r_{i,k} - X_{i,k}\right), \forall i, k \tag{4}$$

$$\tilde{U}_{i,k} = \min\left(T - d_{i,k} + 1, r_{i,k} + X_{i,k}\right), \forall i, k \tag{5}$$

3.4. Scheduling criteria

3.4.1. Peak load

Peak load reduction is the main target of DSM in order to reduce total power demand during peak periods and, hence, enhancing power grid sustainability and reducing overall cost. The peak load is the maximum load over the scheduling time horizon (Hassan et al., 2013). It can be expressed mathematically as in Eq. (6). This mathematical expression can be adopted as scheduling criteria to develop load schedules that minimize the peak load (see Eq. (7)).

$$PL = \max_{t\in[1,T]}\left(N(t) + \sum_{k=1}^{K}\sum_{i=1}^{M_k} e_{i,k}(t)\right) \tag{6}$$

$$\min_u f1 = PL \tag{7}$$

3.4.2. Energy cost

For a given load schedule, the energy cost is usually calculated based on the TOU tariff which is set and communicated by the power grid operator to the consumer. In other words, the energy cost of scheduled loads at any time slot t, $SC(t)$, can be calculated as shown in Eq. (8), where $c(t)$ is the energy unit tariff at time t.

$$SC(t) = SG(t).c(t), \forall t \tag{8}$$

However, at each time slot, it is assumed that the PV units generate an aggregate amount of energy, $PVG(t)$, that saves an amount of energy cost that should be deducted from the total energy cost. In this regard, we attempt to formulate a cost function (net energy cost) that can be utilized to maximize the utilization of the PV output and TOU. As such, the net energy cost that will be paid by the consumer to the grid in each time slot, denoted by $NC(t)$, can be represented as shown in Eq. (9). The total net cost over the scheduling time horizon T (TNC) is calculated as shown in Eq. (10). This function accounts only for the paid cost by the consumers to the grid and neglects the returns from the excess PV output. As such, in the cases that TOU is not applicable, this cost function favors the load schedules that utilizes effectively the PV output. In particular, it can be used as scheduling criteria to develop load schedules that minimize the energy cost paid by the consumers (see Eq. (11)).

$$NC(t) = \begin{cases} (SC(t) - PVG(t)).c(t), & SC(t) - PVG(t) > 0 \\ 0, & SC(t) - PVG(t) \le 0 \end{cases}, \forall t \tag{9}$$

$$TNC = \sum_{t=1}^{T} NC(t) \tag{10}$$

$$\min_u f2 = TNC \tag{11}$$

3.4.3. Consumer inconvenience

The consumer inconvenience can be defined as the deviation of the load schedules from the consumer's preferred schedules. It can be measured from either time or power perspectives (Hassan et al., 2013). From the time perspective, the lower the deviation from preferred starting times of loads, the lower the consumer inconvenience and vice versa. From the power perspective, the lower the deviation of the scheduled amount of power in each time slot from the demanded amount of power, the lower the consumer inconvenience. Similar to Hassan et al. (2013), we consider the power perspective to measure the consumer inconvenience, which can be expressed as the sum of squared deviations of power. The required power by consumer k during time slot t can be computed as shown in Eq. (12). This presents the preferred load schedule for the consumer k.

$$SL_k^{uc}(t) = N_k(t) + \sum_{i=1}^{M_k} e_{i,k}(t), \quad \forall t, k \tag{12}$$

The squared deviation of any schedule generated from k^{th} consumer preferred schedule can be measured as follows:

$$D_k = \sum_{t=1}^{T}(Q_k(t) - SL_k^{uc}(t))^2, \quad \forall k \tag{13}$$

where, $Q_k(t)$ is the total power of re-scheduled loads of consumer k in time slot t based on the shifting value for each flexible load ($u_{i,k}$) (Hassan et al., 2013). Eq. (14) expresses the aggregate consumer inconvenience. This equation can be considered as an objective function (see Eq. (15)) or a problem constraint that can be incorporated in load scheduling models to control the user inconvenience level.

$$D = \sum_{k=1}^{K} D_k \tag{14}$$

$$\min_u f3 = D \tag{15}$$

4. Optimization approach

4.1. Mathematical model

In literature, several optimization models have been developed for the above-described DSM system to optimize load schedules from different perspectives (Esther and Kumar, 2016; Gelazanskas and Gamage, 2014). Most of the existing load-scheduling models consider specific single objective functions to optimize the load-schedules, presenting a shortcoming of the existing models (Foroozandeh et al., 2021). The existing models are mathematically formulated to allow: minimizing peak load, or minimizing energy cost, while considering consumer convenience as a constraint (Adika and Wang, 2014; Della Croce et al., 2017). Moreover, most of the existing models assume the presence of incentives mechanism such as the TOU tariff, and cannot allow the effective integration of DRES. In particular, most of the existing models possess an objective function as in Eq. (11). Therefore, the application of the existing models is restricted to those conditions. However, optimization models that consider combinations of different scheduling criteria (e.g., *f1, f2,* and *f3*) while taking into account the presence of TOU, DRES, or both, can be formulated. Motivated by that, a load scheduling model

that can be applied to optimize load schedules based on different scheduling criteria is proposed. The model is formulated with a weighted sum objective function that includes three scheduling criteria (energy cost, peak load, and consumer inconvenience). Thus, the same model can be applied to optimize load schedules based on different single objective functions by controlling the values of the weights associated with the three scheduling criteria. In addition, the formulated objective function allows the optimization of load schedules considering multiple scheduling criteria based on their weighted sum. It is worth mentioning that there are several multi-objective optimization approaches that can provide better performance than the weighted sum approach (Foroozandeh et al., 2021). However, the comparison of those multi-objective optimization approaches is not within the scope of this paper. Furthermore, the proposed optimization model can allow the effective utilization of TOU, DRES, or both. As such, it can suit the conditions of different countries or communities that seek for energy management systems that can be applied under their particular conditions. The proposed model can be beneficial to minimize the energy cost of the households, residential and commercial buildings in the countries or communities in which TOU is not applicable. It can be applied to optimize load schedules in a way that maximizes the utilization of the power output of DRES, and thus minimizing the imported energy from the grid. The proposed model is adapted from the presented model in Hassan et al. (2013) that provided a single objective load scheduling model that minimizes the peak load only.

The complete mathematical formulation of the proposed load-scheduling model is given below.

Objective function :

$$\min\ f = w_1 f1 + w_2 f2 + w_3 f3 \tag{16}$$

Subject to :

$$\sum_{t=1}^{T} e_{i,k}(t) = d_{i,k}.p_{i,k}, \quad \forall i, k \tag{17}$$

$$\sum_{t=v}^{v+d_{i,k}-1} e_{i,k}(t) = d_{i,k}.p_{i,k}, \quad \forall i, k; v \in \left[\tilde{L}_{i,k}, \tilde{U}_{i,k}\right] \tag{18}$$

$$\tilde{L}_{i,k} = \max\left(1, r_{i,k} - X_{i,k}\right), \quad \forall i, k \tag{19}$$

$$\tilde{U}_{i,k} = \min\left(T - d_{i,k} + 1, r_{i,k} + X_{i,k}\right), \quad \forall i, k \tag{20}$$

$$N(t) + \sum_{k=1}^{K}\sum_{i=1}^{M_k} e_{i,k}(t) \le s(t) + PVG(t), \forall t \tag{21}$$

The objective function (16) is formulated to allow the flexible application of the proposed load scheduling model to optimize load schedules considering different scheduling criteria. In particular, the objective function encompasses three scheduling criteria: peak load (*f1*), energy cost (*f2*), and user inconvenience (*f3*). Furthermore, three weights are assigned to the three scheduling criteria to allow the adaptation of the mathematical model to optimize the load-schedules from different perspectives. The proposed load scheduling model can be integrated with the modern energy management systems to optimize load schedules based on the objectives of different applications. For instance, the model can be applied at consumer side to develop load-schedules that minimize the energy cost. The objective function can be adapted to cope with this goal by adjusting the weights in the objective function f as follows: $w_1 = 0$, $w_2 = 1$, and $w_3 = 0$. In general, the formulated mathematical model can be employed to optimize load-schedules based on the following single objective functions: minimize peak load (min *f1*), minimize energy cost (min *f2*), or minimize consumer inconvenience (min *f3*). In addition, the mathematical model can be adapted to optimize the load schedules based on the following bi-objective functions: minimize peak load and energy cost (min *f1+f2*), minimize peak load and consumer inconvenience (min *f1+f3*), or minimize energy cost and consumer inconvenience (min *f2+f3*). Finally, the mathematical model can be used to optimize the load-schedules based on the three functions (min *f1+f2+f3*). Furthermore, unequal weights can be assigned to control the relative importance of each scheduling criteria. Therefore, the setting of the weights depends on the application's goals and thus it should be set accordingly.

The mathematical model includes the following sets of constraints. Constraints (17) ensure that the sum of power consumption of each flexible load i of consumer k over the scheduling time horizon T is equal to its total power consumption provided by the user. Constraints (18) indicate that flexible load i of consumer k can only be scheduled in consecutive time slots starting from $t = v$ till $v + d_{i,k} - 1$. Moreover, each flexible load can be scheduled in the interval $\left[\tilde{L}_{i,k}, \tilde{U}_{i,k}\right]$. Constraints (19) and (20) refer to the scheduling intervals limits. Constraints (21) ensure that the total energy consumption at time slot t must not exceed the available energy supply a time slot t which includes both the supply limit provided by the grid $s(t)$ and the PV output $PVG(t)$.

4.2. Cuckoo optimization algorithm

Previous research has investigated several metaheuristics to solve the load scheduling problems as indicated in the literature review. A recent study has shown that the cuckoo search optimization algorithm (CSA) performs better than or at least is comparable to the known algorithms such as GA and PSO (Javaid et al., 2017b). In addition, the cuckoo search algorithm has shown a promising performance with other applications (Lakshminarayanan and Kaur, 2018; Laha and Gupta, 2018). Therefore, the CSA algorithm is utilized to solve the above-described load-scheduling model. In addition, an equivalent mixed integer linear programming model for this mathematical model has been formulated as can be found in Appendix A. The MILP-based load scheduling approach will be used for validating the performance of the cuckoo load scheduling approach.

The CSA algorithm is inspired by the breading behavior of cuckoo species combined with Levy Flight behavior of some fruit flies (Yang and Deb, 2009). Some species of Cuckoos have parasitical breading behavior, in which, some Cuckoos dump their eggs in other birds' nests, called "host nests" whose eggs are very similar to cuckoo eggs and thus they use the services of host birds to hatch their own eggs (Yang, 2010). Occasionally, some host birds discover the alien cuckoos' eggs and expel them from host nests. On the other hand, some fruit flies explore their habitat by flying in straight lines then make sudden 90° turns. The CSA algorithm simulates those two social approaches for searching for the optimal solution. In particular, it relies on the following phases: (1) each cuckoo lays one egg only once a time and dump into randomly chosen nest (generated according to Levy distribution); (2) the best nest is selected for next generation; (3) the number of available host nests is fixed, and the egg laid by a cuckoo is discovered by the host bird with a probability $P_a \in [0, 1]$ (Javaid et al., 2017b). The CSA algorithm is expected to provide a comparable performance to other known metaheuristics for two reasons (Javaid et al., 2017b). First, the CSA strikes a balance between exploitation and exploration of new solutions. In addition, the CSA has only two parameters, the population size of host nests, N_{nest}, and P_a. Once N_{nest} is fixed, P_a essentially controls the elitism and the balance of exploitation and exploration (Yang, 2010). The CSA optimization algorithm, with respect to the above-described load scheduling problem, is outlined as follows:

Input Data:

1: Algorithm inputs: N_{nest}, P_a, N_{itr}

2: Problem data: load data (power consumption, duration, preferred time), PV generation, electricity tariff and bounds of the search domain, $\tilde{L}_{i,k}$ *and* $\tilde{U}_{i,k}$

3: Select objective function $\min f$ *as defined in Eq. (16)*

4 : Generate initial population with N_{nest} *solutions*

5: Set l=0

6: ***while*** $l < N_{itr}$ ***do***

7: generate new solutions through random walk (levy flight) and evaluate their finesses with Eq. (16)

8: generate new solutions with discovery rate P_a *and evaluate their finesses with Eq. (16)*

9: keep record of best solution ever

10: Set l = l + 1

11: ***end while***

First, the optimization algorithm read parameters and the load scheduling problem data are loaded. The algorithm reads the loads data and computes the loads consumption, duration and consumer preferred time slot and calculates total energy for appliances. The algorithm reads PV output profile and TOU data (at line 2). Upper and lower scheduling boundaries for each load are calculated based on Eqs. (4)–(5). At line 4, the algorithm generates initial solutions. The algorithm iterates for N_{itr} iterations, counter "*l*" is used as loop counter (at line 5-10). At line 7, the algorithm randomly generates cuckoos by Levy flights and evaluate their finesses and chooses nests randomly. At line 8, the algorithm generates new solutions with probability P_a and compare their fitnesses with Eq. (16). The algorithm stores the best solution and move it to the next iteration at line 9. The loop counter is updated at line 10. The loop iterates for N_{itr} iterations and best solution is moved from one iteration to the next iteration. The algorithm has been implemented in MATLAB.

5. Results and analysis

Several experiments are conducted to validate the performance of the cuckoo load scheduling approach. Moreover, a case study is performed to demonstrate the applicability of the proposed load scheduling model, considering real-life load data. In particular, the real-life load data are collected from an academic building in Egypt in order to illustrate how the proposed load scheduling model can be implemented to develop load schedules that minimize energy cost at commercial-like/governmental facilities in Egypt and in other countries with similar conditions. We also exploit the collected real-life load data to run the other scenarios of load scheduling optimization.

5.1. Validation with MILP-based load scheduling approach

We formulated a mixed integer linear programming (MILP) model for the above-described load scheduling problem. The formulation of the MILP model is presented and described in Appendix A. The MILP model is used only for validating the performance of the cuckoo load scheduling approach in terms of the computational time and solution accuracy. The MILP is not intended as a solution methodology due to the exploding problem sizes and computational time for realistic size problems. It needs excessive computational time and exponentially increasing number of integer variables as the time periods and number of loads increase and we attempt to solve more realistic household problems.

A set of validation experiments are conducting considering four problem variables (factors) with two levels each, and by randomly generating the values of the main model parameters. We have designed an experiment that relies on several problem variables (factors) that affect the problem size in order to evaluate the two load scheduling approaches under a wide range of problem instances. In particular, the experiment relies on the following factors: the number of loads, allowed shifting, time horizon, and availability of DRES. In this experiment, two levels (low and high) are considered for each factor such that the resulted problem instances are diversified. The first factor is the time horizon (T) where the length of the time horizon affects the problem size. The levels of the time horizon are set to 12 Hrs and 24 Hrs since in real applications the load scheduling approaches may be used to schedule loads over 12 Hrs or 24 Hrs. In the MILP formulation, it will be required to represent each load with 12 decision variables for $T = 12$ and 24 decision variables for $T = 24$. Also, other auxiliary variables will be needed to adjust the logic of the mathematical model and the number of related decision variables will be affected by the length of the scheduling time horizon. Moreover, the number of constraints depend on both the time horizon length and the number of time slots. The second factor that will affect the problem size is the number of loads. Therefore, we considered two levels of this factor where the low level is five loads and the high level is twenty loads. This is compatible with the number of loads that are usually adopted in the literature to evaluate the performance of load scheduling approaches (Javaid et al., 2017b; Qayyum et al., 2015). The third factor is the allowed shifting period where the levels of this factor are set based on the length of the time horizon. The levels of this factor are set to "4 periods" and "Open" for $T = 12$, and the levels are set to "8 periods" and "Open" for $T = 24$. The fourth factor is the availability of DRES that can be either available "Yes" on unavailable "No" such that the presence of DRES will add other decision variables and additional constraints that their numbers will depend on the model formulation.

The other model parameters are determined as follows. For all the problem instances with $T = 12$, the required load data are generated from discrete uniform distributions. In particular, the service time is assumed to be uniformly distributed between 1 and 6; the preferred time is also assumed to follow a uniform distribution between 1 and 6, and the rated consumption of each load is uniformly distributed between 100 and 200. The same parameters for the uniform distributions are considered for the problem instances with $T = 24$ except for the preferred time which is considered to be uniformly distributed between 1 and 8.

Table 1
Validation results of the cuckoo load scheduling approach with the MILP solver.

Case #	Factors Levels				Min *f1* (Peak Load)			Min *f2* (Energy Cost)			Min *f1+f2* (Peak Load + Energy Cost)		
	No. of Loads	Time Horizon	Shifting	PV	MILP	CSA	% Gap	MILP	CSA	% Gap	MILP	CSA	% Gap
1	5	12	4	Yes	332	332	0.00%	3261.50	3261.50	0.00%	3705.25	3705.25	0.00%
2	5	12	Open	Yes	332	332	0.00%	3116.75	3116.75	0.00%	3522.00	3522.00	0.00%
3	5	24	8	Yes	189	189	0.00%	1695.00	1695.00	0.00%	2009.63	2009.63	0.00%
4	5	24	Open	Yes	189	189	0.00%	1695.00	1695.00	0.00%	2009.63	2009.63	0.00%
5	20	12	4	Yes	1116	845	−24.28%	10959.75	10959.75	0.00%	12285.75	12285.75	0.00%
6	20	12	Open	Yes	842	823	−2.26%	10535.25	10535.25	0.00%	**	11680.75	NA
7	20	12	Open	No	**	822	NA	11091.50	11091.50	0.00%	12243.25	12245.00	0.01%
8	20	24	8	Yes	368	362	−1.63%	6462.75	7518.75	16.34%	7405.50	7428.13	0.31%
9	20	24	Open	Yes	385	362	−5.97%	5450.88	5450.88	0.00%	**	7030.38	NA
10	20	24	Open	No	366	364	−0.55%	6159.00	6159.00	0.00%	**	7924.00	NA

** No feasible solution obtained within the time limit where time limit for MILP is 1 hr and time limit for CSA is 5 min.

Two different TOU schemes are used for $T = 12$ and $T = 24$ (see Appendix B, Fig. B.1). The PV output for the problem instances with available DRES and $T = 24$ is shown in Fig. B.2 where the first 12 values of the PV output are used for problem instances with T=12. The generated test problems are summarized in Table 1.

The MILP is used for validating the optimality and convergence performance of the Cuckoo algorithm with a time limit under different problem sizes. The MILP solver and cuckoo search optimization algorithm (CSA) are run in MATLAB on a laptop with a processor of Intel(R) Core(TM) i7-8550U CPU @ 1.80 GHz and 16 GB of RAM. The MILP Solver of MATLAB "intlinprog" is adopted to solve the MILP model. This solver relies upon the branch and bound algorithm which can provide optimal solution for MILP models (Anon, 2020). In all the validation experiments, the MILP solver is allowed to run for a time limit of 1 h (3600 s), and the cuckoo optimization algorithm is allowed to run for only a time limit of 5 min (300 s). The validation is conducted for three different modes of the load scheduling model: minimizing peak load, minimizing energy cost, and minimizing both energy cost and peak load. We have elected to run the different three cases for providing an extensive comparison between the MILP and cuckoo load scheduling approaches under different scheduling criteria and for benchmarking other future formulation approaches and solution algorithms. The optimization results are summarized in Table 1.

The optimization results of the peak load (Min *f1*) indicate that the MILP solver is successful to find the optimal solution of the peak load for the problem instances that have small number of loads (5 loads). The MILP solver has needed a very short time that ranges from a fraction of second to a few seconds to obtain the optimal solution for this set of problem instances. However, the MILP solver has failed to obtain the optimal solution within the set time limit (3600 s) for all the cases that include 20 loads. In some cases, the MILP solver could not even obtain a feasible solution within the 3600 s as in Case #7. On the other side, the cuckoo optimization approach shows a promising performance compared to the MILP solver. It has been successful to obtain the same solution as the MILP solver before the time limit (300 s) is elapsed. The percent of solution gap is zero for all problem instances that have 5 loads, indicating that the cuckoo load scheduling approach is successful to obtain the optimal solution within 300 s. It has actually needed a lower time than the MILP solver to reach the optimal solution. Moreover, for the large size problems (20 loads), the cuckoo optimization approach provides promising performance compared to the MILP solver as the % solution gap is in favor of the cuckoo approach. This implies that the cuckoo load scheduling approach is more recommended for optimizing the peak load than the MILP-based load scheduling approach.

The optimization results of the energy cost (Min *f2*) show that the MILP solver has succeeded to obtain the optimal solution for all the tested cases within the time limit. It has needed a very short time to reach the optimal solution except for Case #9 that needed a relatively higher computational time than the other cases. The cuckoo optimization approach is comparable to the MILP solver. It can realize the optimal solution within the set time limit (300 s) for most of the tested cases. The MILP solver has only performed better than the cuckoo approach in Case #8. Overall, the MILP formulation is more recommended for optimizing energy cost.

For the optimization of both the energy cost and peak load (Min *f1+f2*), the results indicate that the cuckoo load scheduling approach performs better than the MILP solver, within the set time limit. Both the MILP solver and the cuckoo load scheduling approaches have succeeded to obtain the optimal solution within a very short time for the small-scale problem instances (5 loads). Although the MILP has shown a better performance than the cuckoo for some cases that include 20 loads (Cases #7-8), it has failed to obtain a feasible solution for the remaining cases (Cases #6, 9, 10) within the time limit. On the other side, the cuckoo load scheduling approach reached a solution within the time limit for these cases. Thus, the cuckoo load scheduling approach is recommended for optimizing both the energy cost and peak load.

5.2. Comparison with previous studies

Two instances of the load-scheduling problem are adapted from the related literature to conduct further validation tests. In particular, the instances are selected based on the availability of complete problem data for conducting the required comparison. As such, it was convenient to adapt our proposed load scheduling model and use their data to make direct comparisons with them. The first problem instance is reported in AboGaleela et al. (2012) that employed a genetic algorithm to show the impact of a DSM load shifting program on both utility and customer in residential areas. In particular, they have formulated two mathematical models with two different objective functions where the first model is formulated to maximize load factor while the second model is formulated to minimize the energy cost. The second problem instance is taken from Vidal et al. (2014) that adopted an evolutionary algorithm to optimize the load schedules, in order to minimize the consumer bill cost. They proposed a load scheduling model that addresses the energy cost through formulating an objective function that minimizes the gap between the target and scheduled load profiles. They consider the target profile as a function of the inverse of the TOU tariff which is provided by the grid. Therefore, we have attempted to compare between the results of our model and the results of Vidal et al. (2014) that formulated another objective function for the same goal.

Table 2
Comparing optimization results with AboGaleela et al. (2012).

Scheduling Metric	Before DSM	AboGaleela et al. (2012)	Cuckoo Optimization Approach
Peak load (kW)	7.7	5.8	5.5
Energy Cost (L.E)	17.0	16.2	16.195

Table 4
Comparing optimization results with Vidal et al. (2014).

	Before DSM	Vidal et al. (2014)	Cuckoo Optimization Approach
Energy cost ($)	2235	2129.20	2091
Peak load (kW)	1897.6	1228.30	1228.30

In the first problem instance, AboGaleela et al. (2012) proposed an optimization approach for small-size problems that can be used for load scheduling at household level. They considered a small number of loads, representing the six common shiftable appliances in Egyptian households, to validate their optimization approach (see Table B.1). They have considered the TOU pricing mechanism shown in Fig. B.3, where the time horizon is set to $T = 24$ with a time slot duration of 1 h. Furthermore, their mathematical model does not consider the presence of DRES.

The cuckoo load scheduling approach has been applied to determine the optimal schedule for the loads presented in Table B.1, considering the TOU in Fig. B.3. The parameters of the cuckoo optimization algorithm are set to: $N_{nest} = 25$, $P_a = 0.25$, and $N_{itr} = 10{,}000$. The optimization results, with respect to the objective of cost minimization (i.e., w_1=0, w_2=1, w_3=0), is presented in Table 2. The results indicate that the cuckoo approach realizes an energy cost of 16.195 L.E which is almost equal to the energy cost achieved by AboGaleela et al. (2012) (see Table 2).

The proposed load scheduling model can be applied to optimize load schedules based on different scheduling criteria, depending on the application's objective (*Min f1, Min f2, Min f3, etc.*). Therefore, considering the problem data of AboGaleela et al. (2012), the model is also applied to find the optimal load schedule for different objectives where there are seven possible scenarios to optimize load schedules (see Table 3). These optimization scenarios are run in order to demonstrate the applicability of the load scheduling model for addressing different objectives in energy management. Moreover, the scenarios are used to compare the difference between the obtained solutions under single and bi/multi-objectives scenarios. The optimization results are provided in Table 3 and the corresponding scheduling profiles are presented in Fig. 2. The results show that the cuckoo load scheduling approach succeeds to reduce the peak load with 29%, from 7.7 kW to 5.5 kW, as per Scenario #1. In Scenario #2, it reduces the energy cost with about 5% from 17.0 L.E to 16.195 L.E. However, it can be observed that running the fourth optimization scenario, in which the optimization approach is adapted to optimize both *f1* and *f2* simultaneously, outperforms the first scenario in which the objective is to minimize *f1* only. In particular, the fourth optimization scenario terminates at a minimum peak load of 5.5 kW while at an energy cost of 16.195 L.E. This indicates that in practical situations when it is required to minimize the peak load, it is recommended to use load scheduling models that consider the minimization of peak load and energy cost simultaneously. The results indicate that the remaining scenarios (5–7) have almost the same output where the optimization algorithm is ineffective and results in load schedules close to the preferred load schedules (Scenario #3). This can be attributed to the characteristics of the consumer inconvenience function (*f3*) that represents a relatively significant portion of the main objective function (*f*).

The second problem instance is taken from Vidal et al. (2014) and includes a larger number of flexible appliances/loads (see Table B.2). The related TOU scheme is presented in Fig. B.4, where the time horizon is set to $T = 24$ with a time slot duration of 1 h. The load profile of the non-shiftable loads is provided in Fig. B.5.

The optimization results for this problem instance, with respect to the minimization of energy cost, are provided in Table 4. The optimization algorithm terminates at an energy cost of $2091 which is lower than the energy cost achieved by Vidal et al. (2014). Furthermore, the optimization approach realizes this lower cost at the same peak load as in Vidal et al. (2014). This can be attributed to the different formulation of the objective function in Vidal et al. (2014) that attempts to minimize the energy cost through minimizing the gap between the target and scheduled load profiles. For the same problem data, the proposed load scheduling model with the cost function in Eq. (16) has provided a better solution in terms of the energy cost.

The optimization results with respect to the other optimization scenarios are summarized in Table 5. The related load scheduling profiles are provided in Fig. 3. Most importantly, it can be observed again that the optimization model in Scenario #4 realizes better solution than Scenario #1. In Scenario #4, the optimization approach achieves a peak load of 1228.3 kW as in Scenario #1 while at a lower energy cost.

6. A case study in Egypt

In order to illustrate the wide applicability of the proposed model, a case study is conducted based on real-life load data collected from an academic building in Egypt. The building includes rooms for teaching staff and researchers, lecture halls, tutorial rooms, services and administrative rooms and laboratories. Accordingly, it contains different types of devices/loads as summarized in Table B.3.

The preferred time for each load is assumed based on the working conditions in the building. Furthermore, we have assumed that the allowable shifting period for each load is four hours and fixed for all loads. The building loads can be scheduled over a time horizon of 12 h ($T = 12$) since the essential activities in the building extends for 12 h; from 7:00 AM to 7:00 PM. The load profile of the building based on these assumptions is shown in Fig. B.6.

In Egypt, energy prices are accounted based on a sliced consumption scheme (Inclining Block Rate) in which each consumption slice is charged a different price. Table B.4 presents the

Table 3
The optimization results of all possible optimization scenarios based on the first problem instance.

Scheduling Metric	Optimization Scenarios						
	Scenario #1	Scenario #2	Scenario #3	Scenario #4	Scenario #5	Scenario #6	Scenario #7
	min *f1*	min *f2*	min *f3*	min *f1+f2*	min *f1+f3*	min *f2+f3*	min *f1+f2+f3*
Peak Load (*f1*)	5.5	5.5	5.5	5.5	5.5	5.5	5.5
Energy Cost (*f2*)	16.8	16.195	16.8	16.195	16.8	16.8	16.8
Consumer Inconvenience (*f3*)	79.72	110.66	79.72	109.26	79.72	79.72	79.72

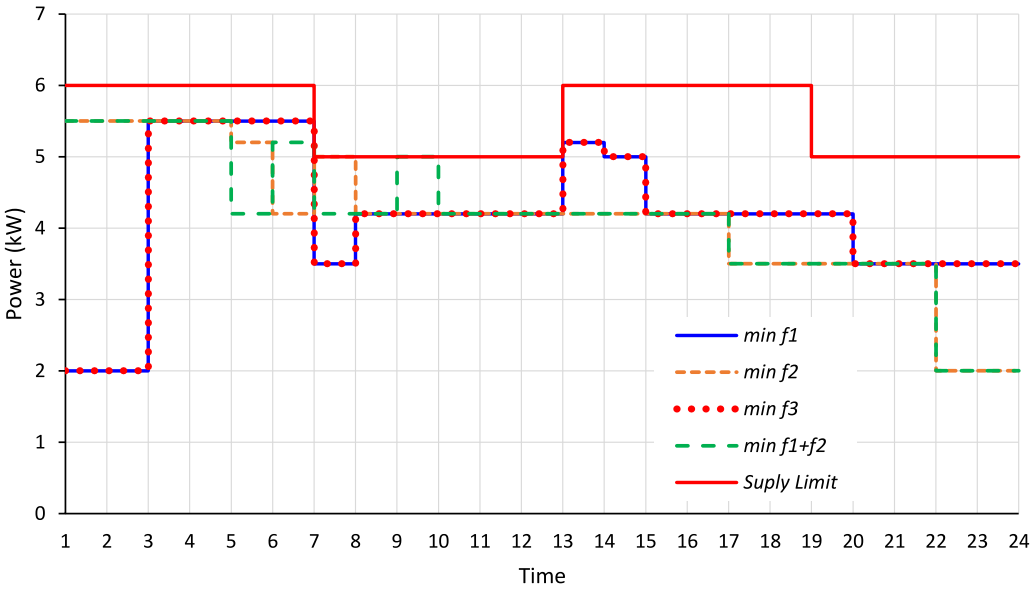


Fig. 2. Scheduling profiles for the first problem instance.

Table 5
The optimization results for all possible optimization scenarios based on the second problem instance.

Scheduling Metric	Optimization Scenarios						
	Scenario #1	Scenario #2	Scenario #3	Scenario #4	Scenario #5	Scenario #6	Scenario #7
	min *f1*	min *f2*	min *f3*	min *f1+f2*	min *f1+f3*	min *f2+f3*	min *f1+f2+f3*
Peak Load (*f1*)	1228.3	1524	1897.6	1228.3	1897.6	1897.6	1897.6
Energy Cost (*f2*)	2130	2091	2235	2092	2236	2236	2236
Consumer Inconvenience (*f3*) $\times 10^3$	1985.5	3651	0	3130	0	0	0

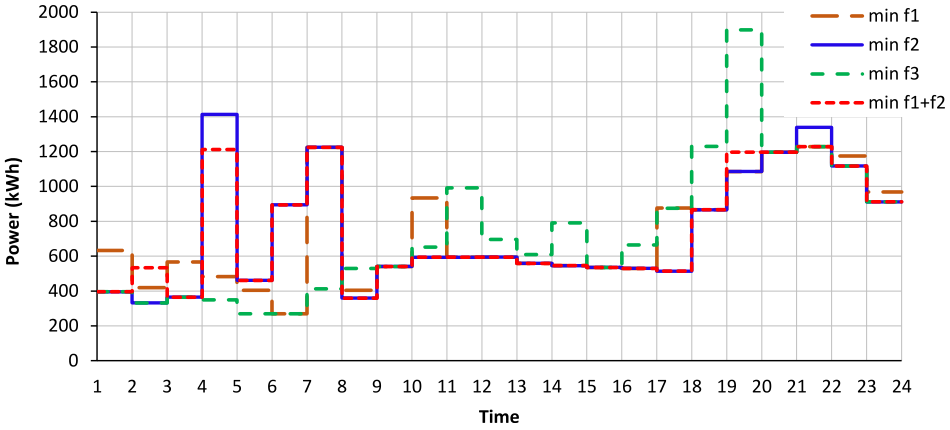


Fig. 3. Scheduling profiles for the second problem instance.

official energy prices per kilowatt-hour in 2018, for commercial buildings. Based on this tariff, the building is charged an estimated energy cost of 35302 L.E per month. We use the cost data in this scheme to estimate the energy cost required to conduct the load scheduling experiments. Two cases are investigated to illustrate the applicability of the proposed load scheduling model under different operating conditions: Fixed Tariff with PV, and TOU with PV. In the first case, it is assumed that the building has an installed PV system, which generates the power output shown in Fig. B.6. This assumption is needed to show how the load scheduling model can be employed in the situations that TOU mechanisms are not applicable, representing the real situation in Egypt. In addition, it is assumed that the energy cost is constant (140 LE per kilowatt-hour) throughout the day. This cost is estimated based on the current consumption pattern in the building and the corresponding consumption slice price in Table B.4. In such situation, the load scheduling approach can be used to develop load schedules that maximize the utilization of the PV output and thus minimizing the energy cost. The PV generation profile can be extended to other DRES through aggregating their power output. Therefore, the applicability of the optimization approach is not confined to PV systems only. In the first case (Fixed Tariff with PV), we consider two instances of the PV generation that differ in the variability of the power output. The first instance (PVG1) is assumed to have a PV generation with a coefficient of variation of 0.31. The second instance (PVG2) is assumed to have a higher variability with a coefficient of variation of 0.40. This will allow to investigate the effectiveness of the proposed load scheduling approach under different operating conditions with respect to PV generation.

Table 6
The optimization results for the case study.

Scheduling Metric	Optimization Scenarios						
	Scenario #1	Scenario #2	Scenario #3	Scenario #4	Scenario #5	Scenario #6	Scenario #7
	min *f1*	min *f2*	min *f3*	min *f1+f2*	min *f1+f3*	min *f2+f3*	min *f1+f2+f3*
The First Case: Fixed Tariff with PV, PVG1 (PV output with low variability)							
Peak Load (*f1*)	105.89	137.36	231.36	108.50	231.23	231.23	231.23
Energy Cost (*f2*)	296.88	255.70	645.40	255.70	645.25	644.59	644.59
Consumer Inconvenience (*f3*)	103143.70	67849.23	0	85762.98	0.033	0.293	0.293
The First Case: Fixed Tariff with PV, PVG2 (PV output with high variability)							
Peak Load (*f1*)	105.89	125.5	231.36	106.6	231.3	231.3	231.23
Energy Cost (*f2*)	491.88	371.22	858.5	381.6	858.3	857.57	857.57
Consumer Inconvenience (*f3*)	123460	108105	0	105929	0.0338	0.2934	0.2934
The Second Case: TOU with PV							
Peak Load (*f1*)	106.3	126.4	231.4	112.3	231.2	231.2	231.2
Energy Cost (*f2*)	227.2	192.1	706.5	193.4	706.3	705.5	705.5
Consumer Inconvenience (*f3*)	100512.2	88124.6	0	89255.7	0.034	0.29	0.29

The results for all the optimization scenarios are provided in Table 6, and the related load profiles are shown in Fig. 4. The results show that the peak load is reduced by 54% for PVG1 and PVG2, i.e. from 231.36 kW in Scenario #3 (preferred consumer's schedule, min *f3*) to 105.89 kW in Scenario #1 that minimizes peak load (min *f1*). In the later scenario, the load profile becomes closer to the PV generation profile; leading to lower energy cost where the energy cost deceases by 54% for PVG1 and 43% for PVG2. In other words, the required energy from the grid is reduced in both instances of PV output. Moreover, the energy cost can be reduced further when running the optimization approach based on Scenario #2 (min *f2*) where it results in a cost reduction of 60% and 57% for PVG1 and PVG2, respectively. However, the fourth scenario (min *f1+f2*) under PVG1 results in the same energy cost as in Scenario # 2 but at a lower peak load. As such, running the fourth optimization scenario provides better solution than Scenario #2 and comparable performance to Scenario #1. The remaining optimization scenarios produce almost the same output of the third scenario. The results indicate that the proposed load scheduling approach will also be effective for the instances of PV generation with high variability.

The second case assumes the existence of TOU tariff and PV. The assumed TOU mechanism is shown in Fig. B.7. The related results are presented in Table 6 and the load profiles are provided in Fig. 5. The results show that the load scheduling approach has successfully utilized the existence of TOU and PV to minimize the energy cost by 80%. The results also confirm that running the fourth optimization scenario provides better solution than Scenario #2 and comparable performance to Scenario #1.

7. Discussion and managerial implications

The results indicate that the proposed load scheduling model, integrated with the cuckoo optimization algorithm, can provide a powerful load scheduling approach for energy management in smart grids. It can be applied to integrate the DRES generation and provide optimal load schedules that minimize the paid energy cost by the consumers. In addition, it can be applied in the situations where both TOU and DRES are present. Finally, it can be employed to optimize load schedules considering different scheduling criteria and under different operating conditions. A case study has been conducted to demonstrate the applicability of the load scheduling model to manage the loads of commercial buildings in the presence of DRES or both TOU and DRES.

From managerial perspectives, the proposed load scheduling approach can be applied for the energy management at different types of consumers: households, residential buildings, and commercial buildings. However, the grid operators should provide the incentives and the energy management tools that facilitate the engagement of the consumers in such DSM programs. For example, most of the countries are transforming to the replacement of traditional power meters with smart meters. Therefore, the proposed load scheduling approach can be integrated in the smart meter's software to allow the consumers to easily control the operations of their loads. The grid operator can also provide online tools to allow the users that do not have smart meters to easily enter the details of their loads and get the proper load schedules for their conditions. This setting is more suitable for application of similar load scheduling approaches in developing countries or in any country with emerging smart grids. The centralization of the energy management systems can allow the grid operators to achieve the objectives of both the consumers and the grid. Overall, the grid operators should devise suitable ways for information exchange with the consumers, and should deploy smart grid infrastructure and energy management systems.

8. Conclusions

Most of the existing load scheduling models consider single scheduling criteria and assume the presence of TOU, and cannot allow the effective integration of DRES. A load scheduling model that can be adapted to optimize load schedules based on different scheduling criteria, taking into account the existence of DRES, TOU or both, is proposed. The cuckoo optimization algorithm is selected to solve the proposed model. Moreover, the equivalent MILP model of the proposed model has been formulated for validating the optimality and computational time performance of the cuckoo load scheduling approach. The validation results indicate that the cuckoo approach, within a small run time limit, provides a promising performance compared to the MILP-based load scheduling approach with respect to the minimization of peak load, and the simultaneous minimization of both peak load and energy cost. However, the MILP-based load scheduling approach is more favored for the optimization of the energy cost. Further validation experiments have shown that the cuckoo approach realizes a better solution than the published results in the literature.

The wide applicability of the proposed load scheduling model has been demonstrated through a case study, which has been conducted based on real-life load data collected from an academic building in Egypt. The results indicate that the buildings under specific conditions can save about 57%–80% of their energy costs through the load scheduling. This demonstrates the applicability of the proposed load scheduling model to allow the effective utilization of DRES (e.g., PV systems) in the situations that TOU mechanisms are not applicable. Therefore, this model can be

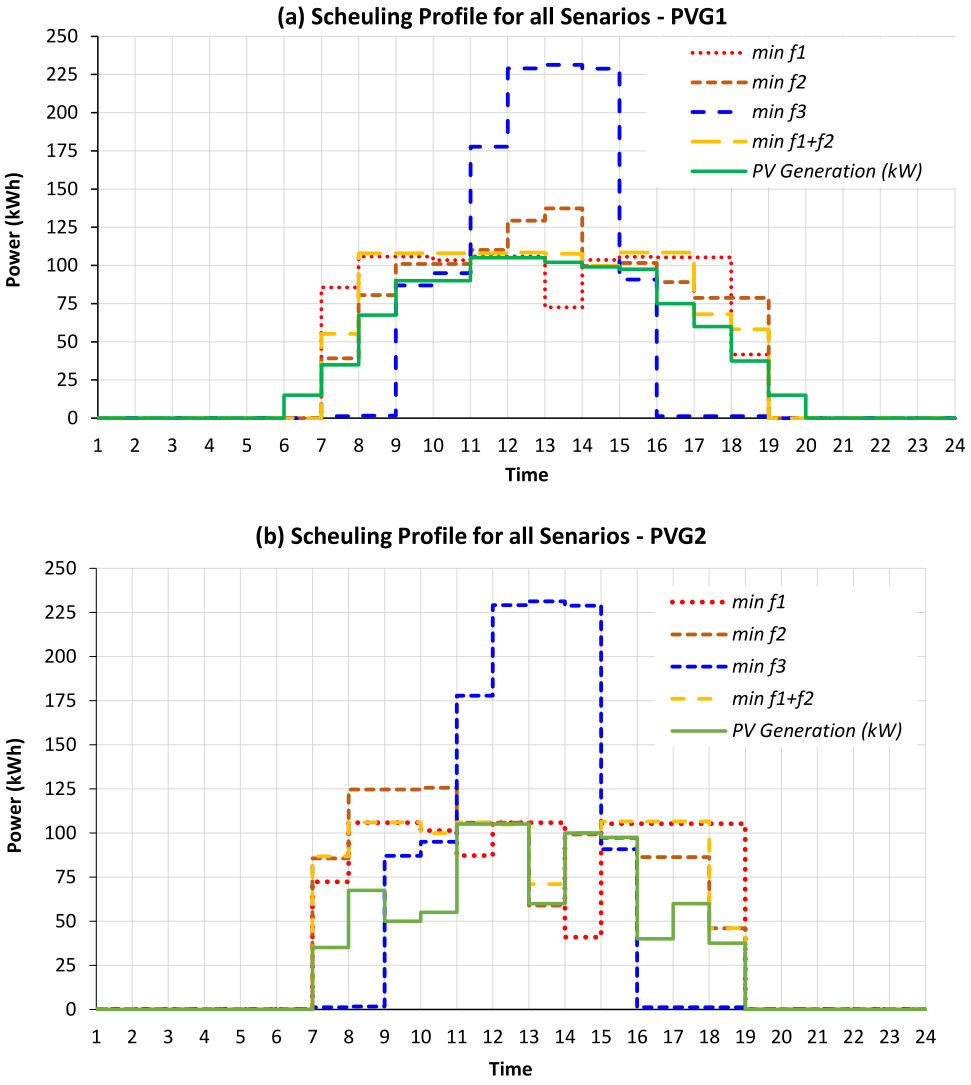


Fig. 4. Scheduling profiles for the first case: Fixed Tariff with PV. (a) Scheduling profiles under PVG1, (b) Scheduling profiles under PVG2.

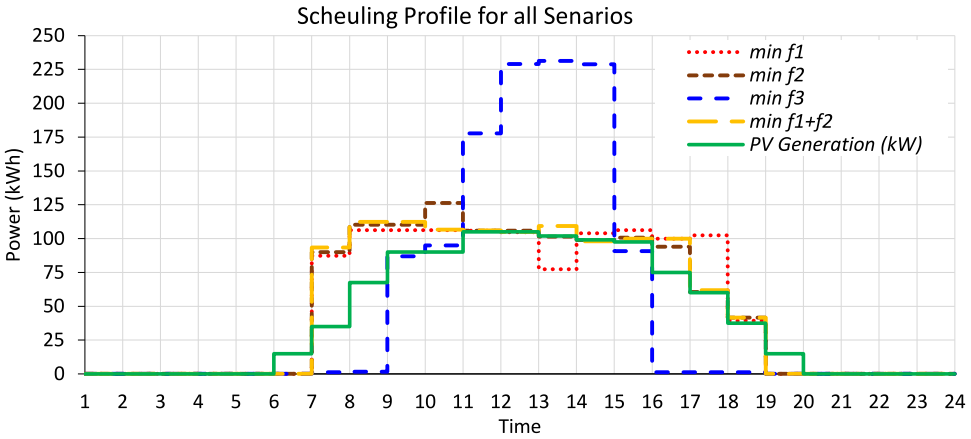


Fig. 5. Scheduling profiles for the second case: TOU with PV.

applied for energy management of residential and commercial buildings in different countries that have either TOU or not. Furthermore, the testing of the different optimization scenarios (bi/multi-objective optimization scenarios) provide another evidence for the wide applicability of the model to serve the objectives of different applications. Moreover, the scenarios results indicate the bi-objective optimization models may provide better solutions than the single objective functions models, implying that the grid may elect to use bi-objective load scheduling models for their energy management systems.

This research possesses some limitations that can be addressed in future work as follows. The model applicability should

be verified through other case studies from different countries. The presented load scheduling model encompasses a weighted sum objective function that consider three scheduling criteria. The comparison of other appropriate multi-objective optimization formulations and approaches for this load scheduling problem should be investigated further. The proposed load scheduling model can be adapted to consider other assumptions that match the conditions of particular countries or communities. Moreover, the model should be enhanced to consider the presence of energy storage and the stochasticity of DRES. Future research should consider the evaluation of other metaheuristics, heuristics and matheuristics approaches.

CRediT authorship contribution statement

Ahmed Shaban: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Supervision, Project administration, Funding acquisition. **Hagag Maher:** Software, Visualization, Validation, Data curation, Writing - original draft, Visualization. **Mahmoud Elbayoumi:** Conceptualization, Software, Data curation, Resources, Writing - original draft, Funding acquisition. **Suzan Abdelhady:** Conceptualization, Software, Data curation, Resources, Writing - original draft, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. MILP model

We have formulated an equivalent mixed integer linear programming (MILP) model for the load scheduling problem described in Section 4. The MILP model is used mainly for validating the performance of the cuckoo optimization approach in terms of the computational time and solution accuracy. The MILP model is adapted and modified from the existing models in the literature (Tan et al., 2016; Della Croce et al., 2017).

The indices, parameters and decision variables of the model are given below.

(1) INDICES & SETS
$i = 1, 2, 3, \ldots, M$ (Load index)
$k = 1, 2, 3, \ldots, K$ (Consumer index where we consider a single user such that $K = 1$)
$t = 1, 2, 3, \ldots, T$ (Time slot index)

(2) MODEL PARAMETERS
$c(t)$: Energy cost in time slot t

$$A_{i,t} = \begin{cases} 1, \text{ if appliance i can be scheduled in time slot t} \\ 0, \text{ otherwise} \end{cases}$$

d_i: Service time of load i
p_i: Power consumption of load i
$PVG(t)$: PV output in each time slot t

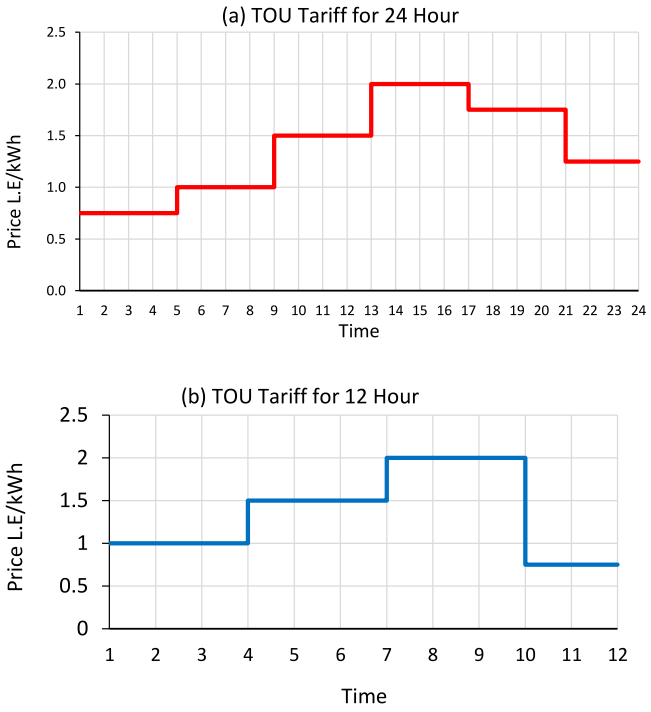


Fig. B.1. TOU Tariffs used in the validation experiments.

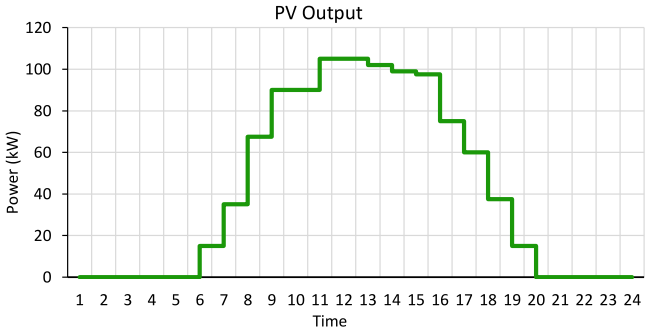


Fig. B.2. PV generation for the validation experiments.

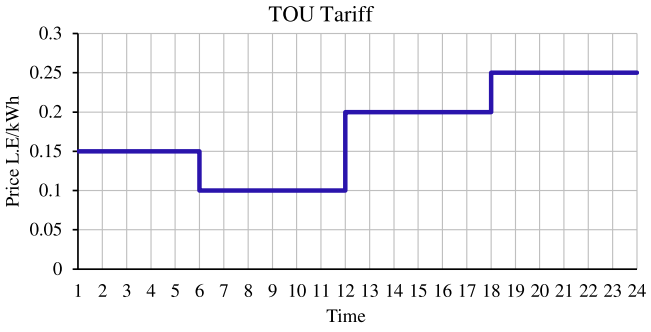


Fig. B.3. TOU Tariff (AboGaleela et al., 2012).

s_t: Supply limit in time slot t.

(3) DECISION VARIABLES

$$x_{i,t} = \begin{cases} 1, \text{ if appliance i is working at time t} \\ 0, \text{ otherwise} \end{cases}$$

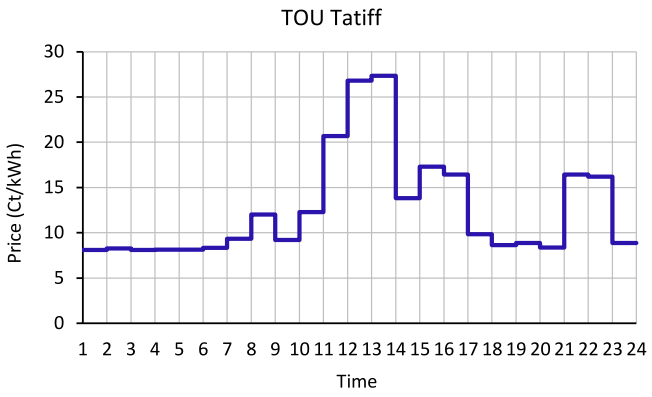


Fig. B.4. TOU Tariff (Vidal et al., 2014).

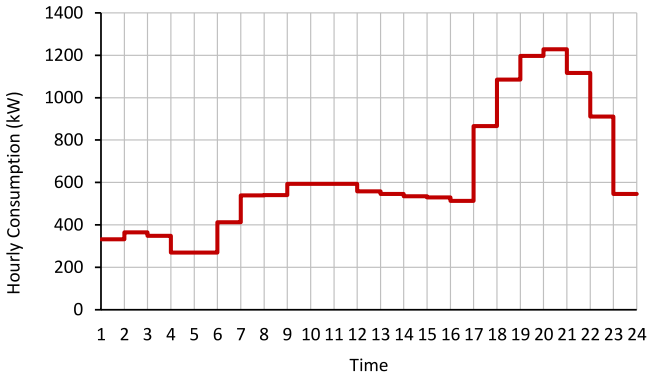


Fig. B.5. Hourly consumption of non-shiftable loads (Vidal et al., 2014).

$$y_{i,t} = \begin{cases} 1, \text{ if appliance i starts working at time t} \\ 0, \text{ otherwise} \end{cases}$$

v_t: Used PV power in time slot t

PL: A continuous decision variable to represent the peak load

The complete MILP model is formulated as follows:

Objective function :

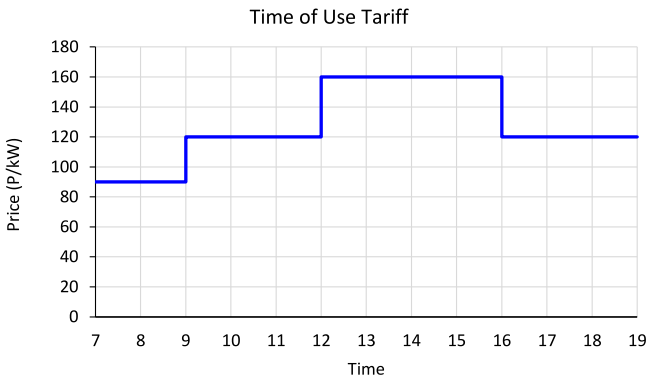


Fig. B.7. Assumed time of use tariff.

Table B.1
Loads data in AboGaleela et al. (2012).

Load i	Appliance Description	Energy Rating (p_i, kW)	Duration (d_i, hr)	Preferred starting time (r_i)
1	Washing machine	0.8	1	19
2	Water heater	0.7	12	6
3	Refrigerator	2.0	24	1
4	Air conditioner	3.5	4	13
5	Microwave	1.0	1	12
6	Water pump	1.5	17	5

$$\min Z = \left(\alpha_1 \left(\sum_{i=1}^{M} \sum_{t=1}^{T} c(t) \cdot [p_i x_{i,t}] - \sum_{t=1}^{T} c(t) \cdot v_t \right) \right) + (\alpha_2 \cdot (PL)) + C \quad (A.1)$$

Subject to :

$$\sum_{i=1}^{M} p_i x_{i,t} \leq s_t + PVG(t), \quad \forall t \quad (A.2)$$

$$v_t \leq PVG(t), \forall t \quad (A.3)$$

$$v_t \leq \sum_{i=1}^{M} p_i x_{i,t}, \quad \forall t \quad (A.4)$$

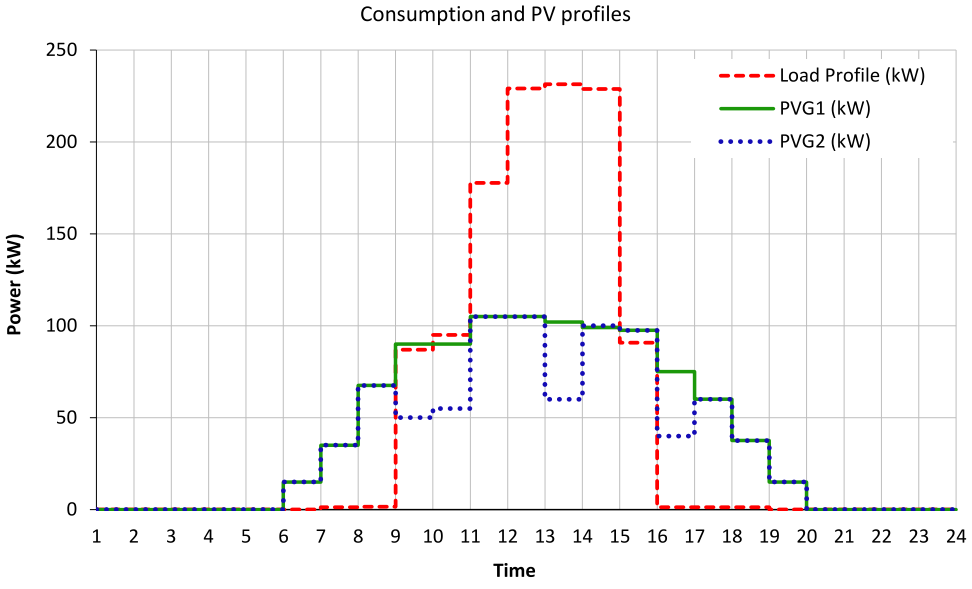


Fig. B.6. Load profile for the selected building based on the preferred time.

Table B.2
Shiftable loads in residential area (Vidal et al., 2014).

Device Type	Quantity	Initial schedule	Hourly consumption (kW)		
			1st hour	2nd hour	3rd hour
Dryer	189	17:00	1.2	0	0
Dish washer	288	14:00	0.7	0	0
Washing machine	268	16:00	0.5	0.4	0
Oven	279	11:00	1.3	0	0
Iron	340	18:00	1.0	0	0
Vacuum cleaner	158	10:00	0.4	0	0
Fan	288	12:00	0.2	0.2	0.2
Kettle	406	19:00	2.0	0	0
Toaster	48	08:00	0.9	0	0
Rice-cooker	59	12:00	0.85	0	0
Hair dryer	58	08:00	1.5	0	0
Blender	66	09:00	0.3	0	0
Frying pan	101	00:00	1.1	0	0
Coffee maker	56	08:00	0.8	0	0
Total	2604	–	–	–	–

$$\alpha_2 \left(\sum_{i=1}^{M} p_i x_{i,t} - PL \right) \leq 0, \quad \forall t \tag{A.5}$$

$$\sum_{t=1}^{T-d_i+1} y_{i,t} = 1, \quad \forall i \tag{A.6}$$

$$\sum_{t=1}^{T} A_{i,t} x_{i,t} = d_i, \quad \forall i \tag{A.7}$$

$$0 \leq \left(\sum_{t=w}^{w+d_i-1} x_{i,t} \right) - d_i y_{i,t} \leq d_i - 1, \forall i, w = 1, 2, 3, \ldots T - d_i + 1 \tag{A.8}$$

$$v_t \geq 0, \forall t \tag{A.9}$$

$$x_{i,t} \in \{0, 1\}, \forall i, t \tag{A.10}$$

$$y_{i,t} \in \{0, 1\}, \forall i, t \tag{A.11}$$

The objective function (A.1) encompasses three scheduling criteria: energy cost, peak load, and user inconvenience. The first part of the objective function represents the energy cost function that can be employed to find load schedules that minimize the energy cost. This cost function is the linear formulation of the cost function in Section 4 (see Eqs. (8)–(11)). In particular, it represents the difference between the total energy consumption cost and the saved cost from the utilization of the PV unit, over the time horizon T. For each time slot t, the energy cost of the scheduled appliances depends on the state of each load at time slot t ($x_{i,t}$), power consumption of each load (p_i), and energy price at time slot t(c(t)). The state of each load is represented by the binary decision variable $x_{i,t}$ where $x_{i,t} = 1$ if the load is active at time slot t and $x_{i,t} = 0$, otherwise. The saved energy cost at each time slot t depends on the used PV output v_t and the energy price c(t). The second part of the objective function (α_2 (PL)) expresses the peak load function which is represented by the continuous decision variable PL. The third part of the objective function is the user inconvenience which is considered as a constant in this model. Furthermore, two weights are assigned for the first two parts of the objective function such that the MILP model can be applied to minimize energy cost by setting $\alpha_1 = 1$ and $\alpha_2 = 0$, to minimize peak load by setting $\alpha_1 = 0$ and $\alpha_2 = 1$, and to minimize both energy cost and peak load by setting $\alpha_1 = 1$ and $\alpha_2 = 1$.

The model's constraints are linearly formulated to represent the set of constraints of the mathematical model presented in Section 4. The first set of constraints (A.2) are required to ensure that the total energy consumption at each time slot t must not exceed the supply limit (s_t) plus the PV output. The total energy consumption depends on the state of the loads at each time slot t and their power consumptions p_i. Constraints (A.3) ensure that the used PV output at time slot t, represented by the continuous decision variable v_t, will not exceed the available PV output at time slot t. Furthermore, constraints (A.4) restrict the value of v_t to be lower than or equal to the total consumption in time slot t. Constraints (A.5) is related to the peak load in the objective function where it enforces the total power consumption to decrease, leading to the minimization of peak load. This constraint will be inactive when $\alpha_2 = 0$. Constraints (A.6) assign a single

Table B.4
Egyptian electricity tariff for commercial users in 2018 (Anon, 2018).

Electricity Selling Price		Customer Service	
Sliced Consumption (kWh/month)	Piasters /kWh	Sliced Consumption (kWh/month)	Customer Service Charge Pound/consumer-month
0 – 100	45	0 – 100	5
101 – 250	84	101 – 250	15
251 – 600	96	251 – 600	20
601 – 1000	135	601 – 1000	25
Zero - More than 1000	140	More than 1000	40
		Zero Read	9

Table B.3
Load data for the case study.

Device Type	Quantity	Power (W)	Preferred Time	Duration	Total Consumption (kWh/day)
2 × 36 Tube	528	72	11	4	152.064
3 × 36 Tube	128	108	12	3	41.472
4 × 36 Tube	232	144	9	6	200.448
6 × 18 Tube	105	180	9	7	132.3
4 × 18 Tube	281	72	9	7	141.624
Saving Lamp	228	30	9	3	20.52
Wall Fans	123	65	10	5	39.975
Ceil Fans	133	75	12	4	39.9
TV	1	130	13	2	0.26
Fridges	4	180	0	24	17.28
PC	101	400	12	4	161.6
Projector	22	275	9	3	18.15
Air-conditioning	20	2238	11	4	179.04
Cooler	3	112	8	5	1.68
Router	2	250	0	24	12
Printer	10	Printing:250 W	13	1	0.5
		Ready:2 W	13	1	0.002
Total	1921				1131.533

starting point for each load i over the time horizon T where $y_{i,t}$ is a binary decision variable that describes the start time of each load i such that $y_{i,t} = 1$ if the load starts at time t and $y_{i,t} = 0$, otherwise. Constraints (A.7) restrict the schedule of each load to be within the feasible time interval where $A_{i,t}$ is a binary parameter such that $A_{i,t} = 1$ if the load can be operated at time t and $A_{i,t} = 0$ for the infeasible time slots. The sum of product of $A_{i,t}$ and $x_{i,t}$ should be equal to the service time of the load to confirm that the loads are run in their feasible times that are determined based on the preferred time for running the load and the allowed shifting period for each load ($X_{i,k}$). The loads are assumed to work continuously without interruption and this assumption is realized through constraints (A.8). Finally, the variables domains are defined in constraints (A.9) through (A.11).

Appendix B. Figures & Tables

See Figs. B.1–B.7 and Tables B.1 and B.4.

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